

Yanxin Luo

Tel: 984-317-4126 | luoyx28@hotmail.com | [LinkedIn](#) | [GitHub](#)

TECHNICAL SKILLS

- **Languages:** C(Advanced), C++(Advanced), Linux C(Advanced), Python(Advanced), shell, HTML, Javascript
- **MCU (Micro Chip Unit):** STM32, Freescale, ESP32
- **System:** Linux, freeRTOS, UCOSII

PROFESSIONAL SUMMARY

Experienced Embedded Software Engineer with 6+ years of experience, specializing in electrical engineering, STM32 MCU, Freescale MCU chips coding, FreeRTOS, UCOSII, Qt and Linux C coding experience. A strong history of leading diverse embedded software engineering projects to deliver next-generation products across drone, instrument, logistics-robot and power industries.

PROJECT

Award of Generative AI Hackathon, Duke University

Durham, NC

AI Language Learning Product "TalkwithMe"

November 2023

- Developed a customized [AI-powered English learning tool](#) using Azure's Text-to-Speech and Pronunciation Assessment APIs.
- Implemented Azure's Text-to-Speech and Pronunciation Assessment APIs for personalized pronunciation feedback in the language learning tool.
- Utilized HTML and JavaScript to create a seamless, user-friendly interface, enhancing the interactive experience for language learners.

WORK EXPERIENCE

Zenitek Intelligent Technology Co., Ltd.

Shanghai, China

Senior Embedded Software Engineer

December 2021 – August 2023

- Participated in the research, development, and debugging of the company's power detection equipment embedded software system. Collaborated across various phases, contributing to a 20% reduction in software bugs, ensuring a more reliable software system.
- Took an active role in developing and verifying key algorithms crucial for the company's power detection technology. Resulted in a 15% improvement in algorithm accuracy and overall system performance.

Shanghai Quicktron Intelligent

Shanghai, China

Senior Embedded Software Engineer

April 2021 – December 2023

- Led the coding, maintenance, and testing of radar calibration code for panels, achieving a remarkable 50% increase in calibration accuracy. Ensured precision and accuracy across the entire fleet of autonomous storage robots.
- Organized and refactored code into modular architectures for stacker trucks, autonomous storage robots, and various panels. Resulted in a substantial 30% reduction in code complexity and enhanced overall code maintainability.
- Managed, maintained, and modified bootloader code for panels and D-box programs, achieving seamless compatibility across all vehicle models. Implemented modifications that led to a 20% reduction in boot time.
- Developed and debugged experimental fixture programs, contributing to the execution of 15% more experiments. Improved program efficiency with a 25% reduction in debugging time.

Focused Photonics Inc.

Hangzhou, China

Junior Embedded Software Engineer

October 2018 – April 2021

- Led the iterative software development and testing for dust monitoring devices, resulting in a 30% reduction in development cycles and an accelerated time-to-market.
- Participated in formulating and maintaining communication protocols using Modbus, achieving a 25% improvement in data transmission efficiency.
- Designed the human-machine interaction logic code for NO₂/SO₂ monitoring station systems, enhancing responsiveness by 20% and improving overall user experience.
- Played a key role in software support, contributing to a 15% decrease in system downtime. Designed effective factory test programs, resulting in a 20% improvement in testing efficiency.

Sunward Intelligent Equipment Co., Ltd.**Changsha, China***Junior Embedded Software Engineer**July 2017 – September 2018*

- Led UAV communication using MDK, achieving a 25% improvement in data transfer speed. Reduced communication latency by 30%, enhancing real-time responsiveness.
- Participated in UAV hardware selection, contributing to a 15% increase in overall system performance. Implemented security enhancements, leading to a 20% reduction in vulnerability risks.
- Conducted rigorous testing to fortify overall system security, reducing potential vulnerabilities by 25%. Ensured a robust and secure unmanned vehicle system.
- Directed wireless CAN bus links for seamless data exchange in an intelligent unmanned vehicle project. Implemented protocols for a 20% increase in data transfer efficiency, ensuring a secure and efficient integration.

EDUCATION**Hengyang Normal University****Hengyang, China***Bachelor of Applied Science: Electronic Information Engineering**September 2013 – July 2017*

- Relevant Courses: Digital Circuit Design | Analog Circuit Principles | Signal Analysis and Processing | Linux and C-Programming | Data Structures and Algorithms in C